

Problem

A customer in Victoria, Australia is considering installing solar PV and a battery at an office building they own. They hope to save money on their energy costs. We want to evaluate their expected annual savings.

The customer's energy charges are made up of two main components:

- A network tariff with independent weekday and weekend demand charges. The network tariff charges \$12/kW for the maximum power imported from the grid each month on weekdays and \$3/kW for the same quantity on weekends. Power is calculated using energy import values and averaging at the half-hour level.
- A wholesale energy cost with different \$/kWh rates for each interval and different rates for import from and export to the grid. Rates can be negative whereupon the customer will be paid to import energy from the grid and will have to pay to export energy to the grid.

The customer has externally modelled solar PV data and wishes to model a 500 kWh battery with a power of 250 kW.

Data sources

- `load_data.csv` contains a year of historical load data for the office building:
 Datetime column (Australia/Melbourne)
 Existing kWh load `ImportkWh`
- `solar_data.csv` contains a year of modelled solar PV data:
 Datetime column (GMT+10)
 Solar PV generation in kWh `Generation`
- `market_data.csv` contains forecast wholesale prices for 2024:
 Datetime column (GMT+10)
 Wholesale energy prices `ImportWholesalePrice` and `ExportWholesalePrice` in \$/MWh

Datetime always denotes the start of a half-hour interval. Note that not all data sources cover the same period and will need to be extrapolated somehow. However, statistical modelling is not the aim of this exercise. Each dataset can be considered fully representative of a single year.

Outputs

We would like to see the expected returns for this project for the year 2024. Cashflows should be broken down by component and month. We would also like to see half-hourly interval data showing the battery's optimised behaviour and running state-of-charge values. The battery should be optimised by constructing and solving a (mixed integer) linear programming problem and we will also discuss how the solution has been verified.